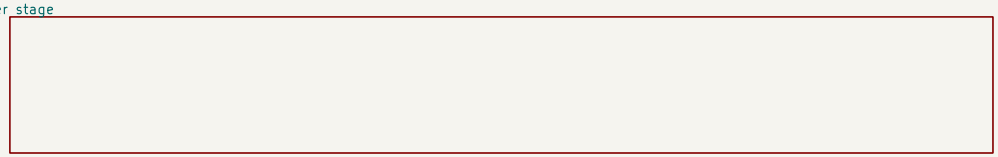
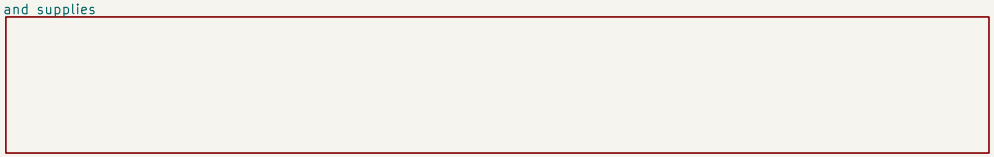
Power stage



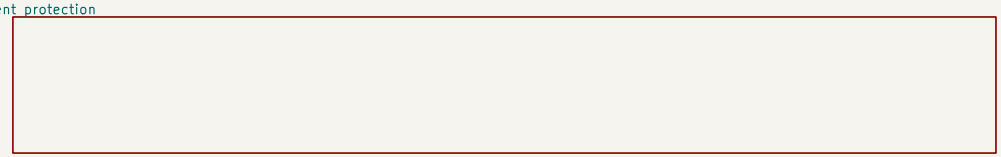
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Gate driver and supplies



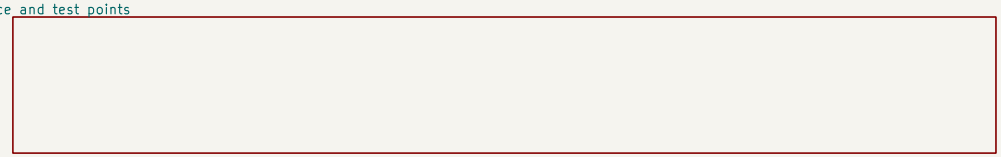
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Independent protection



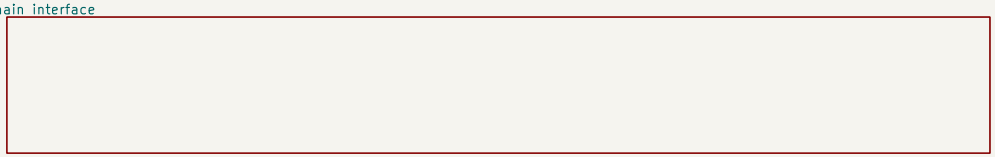
File: buck_3.kicad_sch

STM32 interface and test points



File: buck_4.kicad_sch

Power-domain interface



File: buck_5.kicad_sch

UNAL Manizales

Sheet: /
File: buck.kicad_sch

Title: Buck 12V / 6V – REV B PROTOTYPE

Size: A4	Date:	**Rev: B**
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J1

https://www.mesk...02-500.pdf

12V IN TB002-500-02BE

Q1

https://www.infineon.com/...irfz44n-datasheet-en.pdf

IRFZ44NPBF

D1

SS34

L1

https://www.mura...power-magnetics (KMP_6000B_B01)

680uH 60B684C

RS1

0.050R 1% 2W RLP25FEER050

J2

https://www.mesk...02-500.pdf

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C1

100uF 25V

C2

100uF 25V

C3

100nF 50V

C4

10uF 50V 50SVF10M

C5

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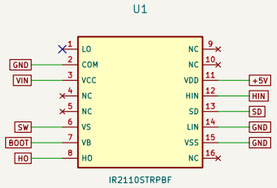
UNAL Manizales – Sistemas Dinamicos y Control

Sheet: /Power stage/

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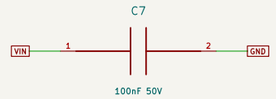
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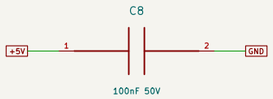
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KiCad E.D.A. 10.0.7		Id: 2/6

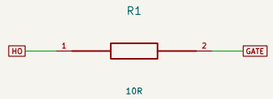


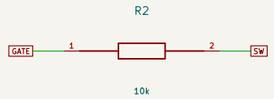


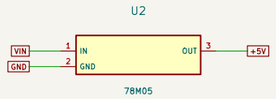


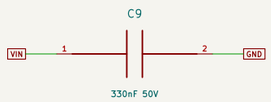




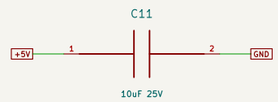


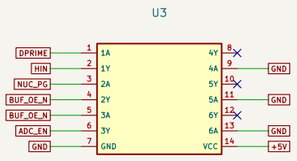


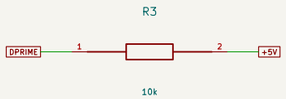


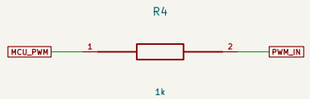


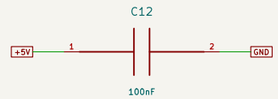










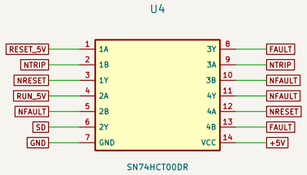


UNAL Manizales – Sistemas Dinamicos y Control

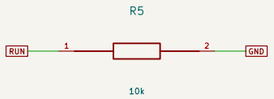
Sheet: /Gate driver and supplies/
File: buck_2.kicad_sch

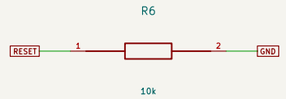
Title: Gate driver and supplies – REV B PROTOTYPE

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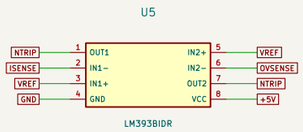




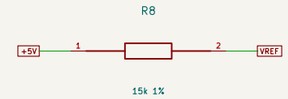


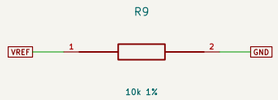


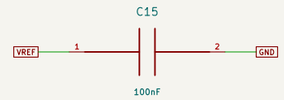




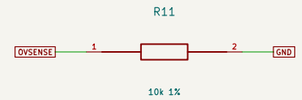


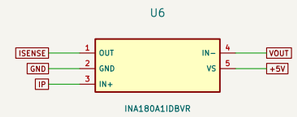




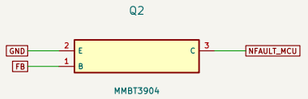


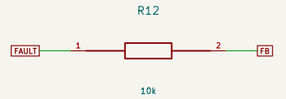


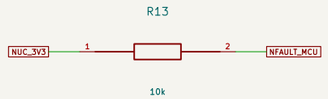












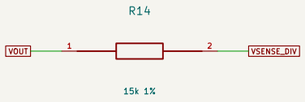
UNAL Manizales – Sistemas Dinamicos y Control

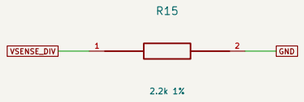
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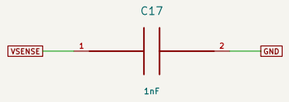
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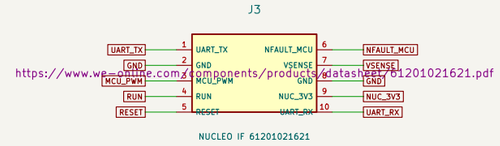
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KiCad E.D.A. 10.0.7		Id: 4/6

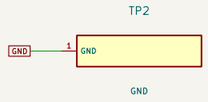


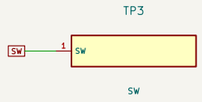


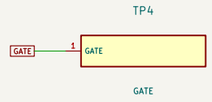


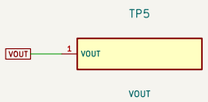


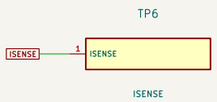


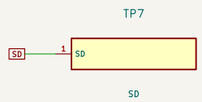


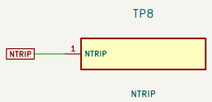


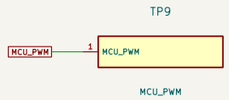










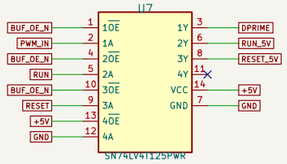


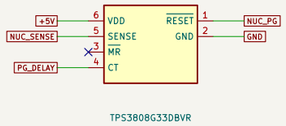
UNAL Manizales – Sistemas Dinamicos y Control

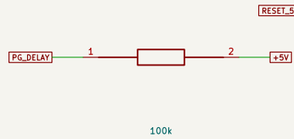
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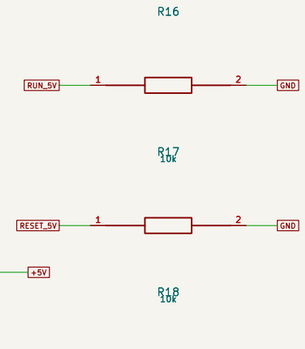
Title: STM32 interface and test points – REV B PROTOTYPE

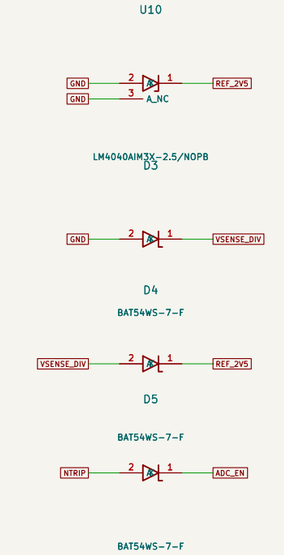
Size: A3	Date:	**Rev: B**
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Rev B power-domain interface. U7 buffers PWM/RUN/RESET from the NUCLEO 3.3 V domain into the Buck 5 V logic; outputs are high-impedance unless U9 sees NUC_3V3 >= 3.07 V (BUF_OE_N via spare U3 gate). R3 keeps DPRIME high (switch off), R16/R

UNAL Manizales – Sistemas Dinamicos y Control		
Sheet: /Power-domain interface/		
File: buck_5.kicad_sch		
Title: Power-domain interface – REV B PROTOTYPE		
Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 6/6